Fiscal Year:	FY 2013	**Task Last Updated:**	FY 08/20/2013
PI Name:	Ferl, Robert Ph.D.		
Project Title:	Molecular Biology of Growth and Cell Remodeling within the Spaceflight Environment		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	32611-0690	**Congressional District:**	3
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2011 Space Biology NNH11ZTT002N
Start Date:	08/01/2012	**End Date:**	12/31/2015
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA KSC
Contact Monitor:	Levine, Howard	**Contact Phone:**	321-861-3502
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date is now 12/31/2014 per NSSC information; original end date 7/31/13 (Ed., 6/24/14) NOTE: End date is now 6/30/2014 per NSSC information; original end date 7/31/13 (Ed., 8/9/13)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Paul, Anna-Lisa (University of Florida)		
Grant/Contract No.:	NNX12AN69G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Plants experiencing spaceflight are quite normal in appearance but can exhibit growth habits distinctly different from plants on Earth. Historically the spaceflight-induced differences were difficult to dissect due to changing hardware and flight conditions. With the Advanced Biological Research System (ABRS) and a consistent ISS orbital environment, it is now possible to confidently approach a dissection of the molecular biology guiding the altered growth of plants in spaceflight. This proposal involves experiments that specifically address growth and molecular changes in Arabidopsis that occur during spaceflight, bringing the molecular and genetic tools of Arabidopsis to bear on fundamental questions of root morphology, growth, and cell wall remodeling. During the APEX-TAGES experiments, we observed a remarkable number of gene expression changes during spaceflight that are associated with cell wall restructuring, especially in roots. We also observed, with the ABRS/GIS imaging system, genetically dependent changes in root growth direction, movement, and structure. The current experiments will employ the ABRS and the GIS to dissect and better understand the molecular changes guiding growth restructuring on orbit. In particular we hypothesize that spaceflight conditions engender a constant state of root cell wall modification during the cell file rotations that characterize spaceflight-induced root skewing. The specific experiments will involve mutants in root growth and cell wall signaling pathways identified in our previous spaceflight molecular studies, as well as new Green Fluorescent Protein biosensor lines also based on the genes identified in the previous flight experiments. The overall goal of our program is to understand fundamental molecular biological responses to extraterrestrial environments. This goal aligns with Fundamental Space Biology goals to understand the impact of spaceflight on biological systems to better enable the exploration imperative. We have used molecular biology and genetically tagged plants as biological monitors of spaceflight and space-related environments with great science return, leading to the development of new biological and hardware tools to study the spaceflight response with even higher fidelity in a wider variety of spaceflight-related environments. The focus of the current proposal is to build on these data and other insights gained from the execution of the APEX-TAGES experiments in the ABRS /GIS, advancing science while also advancing the telemetric imaging hardware and biological experiment support for the ISS and future platforms Gaining insights into such mechanisms is recognized as fundamental within the decadal study and underpins answers to some of the biggest questions in spaceflight plant biology.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	To gain fundamental insight into how plants perceive and process the signals that control tropism and adaptive physiology in novel and extreme environments. It also highlights how the unique research environment of the ISS can provide can address biological questions that cannot be answered on Earth where gravity would mask many of the underlying phenomena.
Task Progress:	Plants experiencing spaceflight are quite normal in appearance but can exhibit growth habits distinctly different from plants on Earth. Historically the spaceflight-induced differences were difficult to dissect due to changing hardware and flight conditions. With the Advanced Biological Research System (ABRS) and a consistent ISS orbital environment, it is now possible to confidently approach a dissection of the molecular biology guiding the altered growth of plants in spaceflight. During the APEX01-TAGES experiments, we observed a remarkable number of gene expression changes during spaceflight that are associated with cell wall restructuring, especially in roots. We also observed, with the ABRS/GIS imaging system, genetically dependent changes in root growth direction, movement and structure. The current experiments will employ the ABRS and the GIS to dissect and better understand the molecular changes guiding growth restructuring on orbit. In particular we hypothesize that spaceflight conditions engender a constant state of root cell wall modification during the cell file rotations that characterize spaceflight-induced root skewing. The specific experiments will involve mutants in root growth and cell wall signaling pathways identified in our previous spaceflight molecular studies, as well as new Green Fluorescent Protein biosensor lines also based on the genes identified in the previous flight experiments. A number of the morphometric phenomenon observed on orbit in APEX-01 can be mimicked on the ground in specialized growth configurations. Pains were taken to establish standardized growing systems that could deliver consistent biology to run parallel studies to NASA flight and comparable ground control studies. On orbit, the flight plants exhibited skewing and waving behavior that was previously thought to be a gravity-dependent phenomenon (Paul et al. 2012, BMC Plant Biol, 12: 232) and we correlated this to aspects of gene expression patterns on orbit (Paul et al. 2013 BMC Plant Biol.13: 112) . Waving and skewing can be easily reproduced in the terrestrial laboratory by growing plants along the surface of agar plates tilted to a 45° slant. We are using this growth set-up to screen mutants, characterize reporter genes, and run experiments that mimic other environmental conditions that could be encountered in an orbital experiment, and also to characterize seed lines being targeted for flight applications in 2014. In addition, a variety of new mutant seed lines and gene reporters chosen on the basis of the previous flight experiment results are being evaluated in the ground studies of the first year of the grant. Several genes which were highly induced in spaceflight were chosen for mutation experiments to characterize the effects the deletion had on terrestrial growth patterns. Of special interest are genes involved in touch-wound-remodeling of cell walls, which are among the highest differentially expressed genes. There is also likely a strong connection between the skewing phenomenon seen on orbit and this group of genes. A number of new reporter gene lines are under development as well, currently 17, which are in various stages of completion to fully homozygous, usable lines.
Bibliography Type:	Description: (Last Updated: 11/27/2022)
Articles in Peer-reviewed Journals	Schultz ER, Kelley KL, Paul A-L, Ferl RJ. "A method for preparing spaceflight RNAlater-fixed Arabidopsis thaliana (Brassicaceae) tissue for scanning electron microscopy." Applications in Plant Sciences. 2013 Aug;1(8):1300034. http://dx.doi.org/10.3732/apps.1300034 , Aug-2013